

(1) 由以上信息可以得知表格如下，其中交换机A作为根交换机。

col1	交换机A	交换机B
Priority	4096	32768
BridgeAddr	5869.6c15.59e2	5869.6c15.59ca
DesignatedRoot	4096.5869.6c15.59e2	4096.5869.6c15.59e2
RootCost	0	20000
RootPort	-	0/1
Designated	0/1 0/2	-

(2) 此时若将端口0/1之间的链路down掉，则交换机B的0/2端口由阻塞状态转变为转发状态，首先在拔掉之前运行命令查看交换机B的0/2端口状态，此时仍然是discarding，在切断端口0/1链路后，经过2秒时间后（计时器计时）再次查看交换机B的0/2端口状态，发现已经改变变为forwarding，整体用时2s左右：

```

PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode : None
PortState : discarding
PortPriority : 128
PortDesignatedRoot : 4096.5869.6c15.59e2
PortDesignatedCost : 0
PortDesignatedBridge : 4096.5869.6c15.59e2
PortDesignatedPortPriority : 128
PortDesignatedPort : 2
PortForwardTransitions : 7
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : alternatePort
20-S5750-2(config)#*Oct 24 03:03:20: %SPANTREE-5-ROOTCHANGE: Root Changed: New Root Po
rt is GigabitEthernet 0/2. New Root Mac Address is 5869.6c15.59e2.
*Oct 24 03:03:21: %SPANTRREE-5-TOPOTRAP: Topology Change Trap.
*Oct 24 03:03:22: %LINK-3-UPDOWN: Interface GigabitEthernet 0/1, changed state to down
.
*Oct 24 03:03:22: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet 0/1,
changed state to down.
show spanning-tree interface gigabitEthernet 0/2

PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFilter : Disabled
PortGuardmode : None
PortState : forwarding
PortPriority : 128
PortDesignatedRoot : 4096.5869.6c15.59e2
PortDesignatedCost : 0
PortDesignatedBridge : 4096.5869.6c15.59e2
PortDesignatedPortPriority : 128
PortDesignatedPort : 2
PortForwardTransitions : 8
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : rootPort
20-S5750-2(config)#

```

(3) 此时交换机的信息如下:

交换机A:

```

20-S5750-1(config)#show spanning-tree
StpVersion : RSTP
SysStpStatus : ENABLED
MaxAge : 20
HelloTime : 2
ForwardDelay : 15
BridgeMaxAge : 20
BridgeHelloTime : 2
BridgeForwardDelay : 15
MaxHops: 20
TxHoldCount : 3
PathCostMethod : Long
BPDUGuard : Disabled
BPDUFilter : Disabled
LoopGuardDef : Disabled
BridgeAddr : 5869.6c15.59e2
Priority: 4096
TimeSinceTopologyChange : 0d:0h:2m:5s
TopologyChanges : 3
DesignatedRoot : 4096.5869.6c15.59e2
RootCost : 0
RootPort : 0
20-S5750-1(config)#

```

交换机B:

```

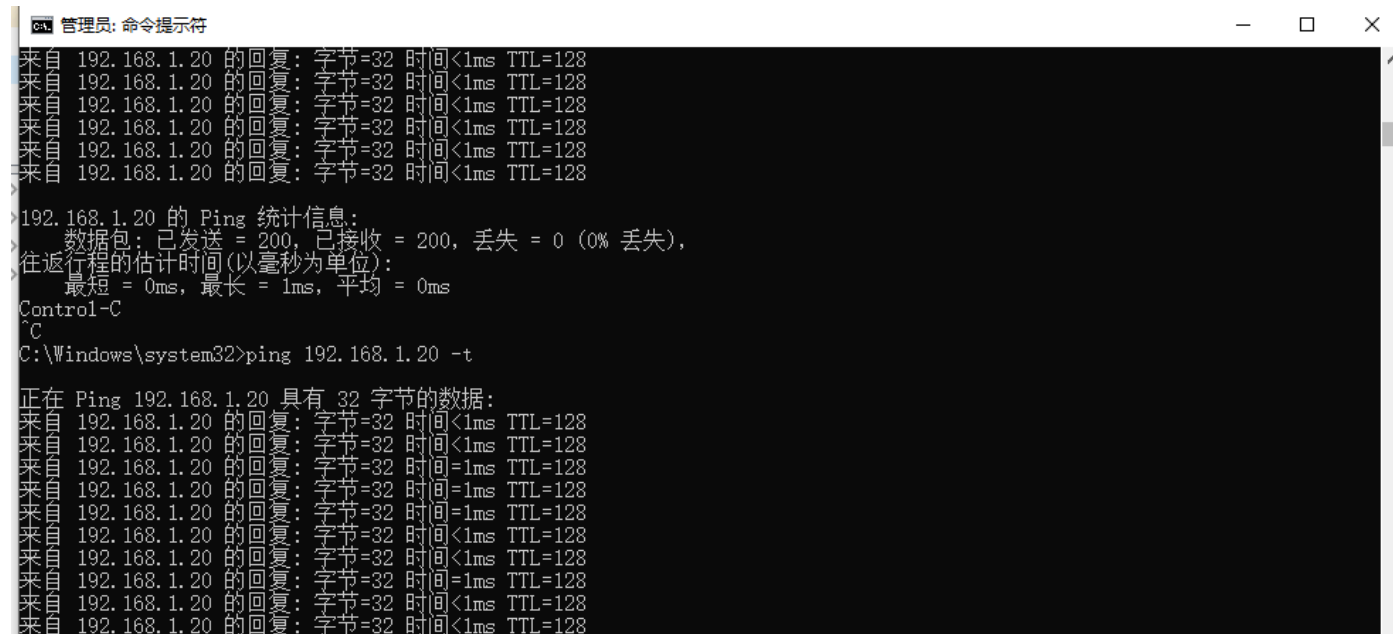
20-S5750-2(config)#show spanning-tree
StpVersion : RSTP
SysStpStatus : ENABLED
MaxAge : 20
HelloTime : 2
ForwardDelay : 15
BridgeMaxAge : 20
BridgeHelloTime : 2
BridgeForwardDelay : 15
MaxHops: 20
TxHoldCount : 3
PathCostMethod : Long
BPDUGuard : Disabled
BPDUFilter : Disabled
LoopGuardDef : Disabled
BridgeAddr : 5869.6c15.59ca
Priority: 32768
TimeSinceTopologyChange : 0d:0h:2m:44s
TopologyChanges : 4
DesignatedRoot : 4096.5869.6c15.59e2
RootCost : 20000
RootPort : GigabitEthernet 0/2
20-S5750-2(config)#

```

col1	交换机A	交换机B
Priority	4096	32768
BridgeAddr	5869.6c15.59e2	5869.6c15.59ca
DesignatedRoot	4096.5869.6c15.59e2	4096.5869.6c15.59e2
RootCost	0	20000
RootPort	-	0/2
Designated	0/1 0/2	-

与（1）进行比较，可以发现交换机B的根端口已经转变为了端口0/2，说明在端口0/1链路断开后，交换机会重新计算生成树，动态调整使得端口0/2处于打开状态，确保新的拓扑结构能够稳定下来，防止环路。

（4）将网络拓扑恢复到没有断开端口0/1之前的状态，此时交换机B是非根交换机，端口0/1是根端口，端口0/2是阻塞端口。然后执行ping命令，并在过程中拔掉交换机A与交换机B端口0/1之间的连线，发现并没有丢包产生：



```
管理员: 命令提示符
来自 192.168.1.20 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间<1ms TTL=128
> 192.168.1.20 的 Ping 统计信息:
    数据包: 已发送 = 200, 已接收 = 200, 丢失 = 0 (0% 丢失),
    往返行程的估计时间(以毫秒为单位):
        最短 = 0ms, 最长 = 1ms, 平均 = 0ms
Control-C
^C
C:\Windows\system32>ping 192.168.1.20 -t

正在 Ping 192.168.1.20 具有 32 字节的数据:
来自 192.168.1.20 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间=1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间=1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间=1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间=1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间<1ms TTL=128
来自 192.168.1.20 的回复: 字节=32 时间<1ms TTL=128
```

过程中交换机的输出如下，从图中可以观察到，在这个过程中，端口0/2从阻塞端口（替换端口）变为了根端口：

```

User sessions
Inconsistent states : normal
PortRole : rootPort
20-S5750-2(config)#show spanning-tree interface gigabitethernet 0/2

PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFILTER : Disabled
PortGuardmode : None
PortState : discarding
PortPriority : 128
PortDesignatedRoot : 4096.5869.6c15.59e2
PortDesignatedCost : 0
PortDesignatedBridge : 4096.5869.6c15.59e2
PortDesignatedPortPriority : 128
PortDesignatedPort : 2
PortForwardTransitions : 1
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : alternatePort
20-S5750-2(config)#Oct 24 02:31:51: %SPANTREE-5-ROOTCHANGE: Root Changed: New Root Port is GigabitEthernet 0/2. New Root Mac Address is 5869.6c15.59e2.
*Oct 24 02:31:51: %SPANTREE-5-TOPOTRAP: Topology Change Trap.
*Oct 24 02:31:53: %LINK-3-UPDOWN: Interface GigabitEthernet 0/1, changed state to down.
*Oct 24 02:31:53: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet 0/1, changed state to down.
*Oct 24 02:31:53: %SPANTREE-5-EVENT: The device has been selected as the Root Bridge.
*Oct 24 02:31:54: %SPANTREE-5-TOPOTRAP: Topology Change Trap.
*Oct 24 02:31:55: %LINK-3-UPDOWN: Interface GigabitEthernet 0/2, changed state to down.
*Oct 24 02:31:55: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet 0/2, changed state to down.
*Oct 24 02:32:01: %SPANTREE-5-ROOTCHANGE: Root Changed: New Root Port is GigabitEthernet 0/2. New Root Mac Address is 5869.6c15.59e2.
*Oct 24 02:32:02: %LINK-3-UPDOWN: Interface GigabitEthernet 0/2, changed state to up.
*Oct 24 02:32:02: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet 0/2, changed state to up.
*Oct 24 02:32:02: %SPANTREE-5-TOPOTRAP: Topology Change Trap.
*Oct 24 02:32:03: %SPANTREE-6-RCVDTCPDU: Received tc bpd on port GigabitEthernet 0/2 on MST0.

20-S5750-2(config)#show spanning-tree interface gigabitethernet 0/2

PortAdminPortFast : Disabled
PortOperPortFast : Disabled
PortAdminAutoEdge : Enabled
PortOperAutoEdge : Disabled
PortAdminLinkType : auto
PortOperLinkType : point-to-point
PortBPDUGuard : Disabled
PortBPDUFILTER : Disabled
PortGuardmode : None
PortState : forwarding
PortPriority : 128
PortDesignatedRoot : 4096.5869.6c15.59e2
PortDesignatedCost : 0
PortDesignatedBridge : 4096.5869.6c15.59e2
PortDesignatedPortPriority : 128
PortDesignatedPort : 2
PortForwardTransitions : 3
PortAdminPathCost : 20000
PortOperPathCost : 20000
Inconsistent states : normal
PortRole : rootPort
20-S5750-2(config)#
```

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正是端口0/2从替换端口转换为了根端口，承担起了数据转发的任务，才使得网络连接正常，没有丢包现象的发生。

(5) 记录表格如下：

col1	交换机A	交换机B
Priority	4096	32768
BridgeAddr	5869.6c15.59e2	5869.6c15.59ca
DesignatedRoot	4096.5869.6c15.59e2	4096.5869.6c15.59e2
RootCost	0	20000
RootPort	-	0/2
Alternate	-	-

同（1）表格进行比较可以发现，交换机B的根端口变成了端口0/2，原因分析同（3）。

注意，由于交换机B上的端口0/1是断开连接了，因此它不能起到替换端口在必要时迅速切换为转发状态的端口的作用，而是被交换机直接关闭了，因此它不属于交换端口，交换端口必须要求端口是连接状态，只是被阻塞了而已。